

KOFvision

KOFvision is a standalone Python viewer for KOF survey files with:

- 2D and 3D visualization
- KOF polyline parsing (09 91 ... 09 99 / 09 96)
- surface triangulation (Delaunay)
- cut/fill volume calculation between two surfaces
- interactive section drawing and profile charts

The main app is implemented in `kof_viewer.py`.

Features

- Open one or more `.kof` files
- Plot points/polylines in 2D plan view
- Plot points/polylines in 3D with optional triangulated surface
- Toggle labels, markers, lines, triangulation, and equal aspect mode
- Load a georeferenced orthophoto/raster background in the 2D tab
- Toggle file layers and raster layer from the Layers panel
- Zoom 2D view to selected (visible) layers
- Select top and bottom surfaces and compute:
 - net volume
 - cut volume
 - fill volume
 - overlap area
- Draw one or more section lines in 2D and open section windows with:
 - top and bottom profiles
 - shaded cut/fill regions
 - net/cut/fill cross-section area

Requirements

- Python 3.10+
- `numpy`
- `matplotlib`
- `scipy`
- `tkinter` (usually included with standard Python installers)

Install Python dependencies:

```
pip install numpy matplotlib scipy
```

Run

Start empty and load files from the UI:

```
python kof_viewer.py
```

Or preload one or more KOF files:

```
python kof_viewer.py "12_A (1).kof" "12_A(1) (1).kof"
```

Workflow

1. Open two surface files (for example existing top and design/bottom).
2. Select Top and Bottom in the second toolbar row.
3. Choose volume Method (**raster** or **triangles**).
4. Optionally enable Triangulation.
5. Click Compute volume.
6. Read net/cut/fill in the toolbar summary.
7. Click Draw section and pick two points in 2D to open a section profile window.

View Controls

- Zoom to selected layers: fit the 2D view to visible layers.
- Reset view: clear manual zoom and return to auto extents.

Raster Background (Orthophoto)

The 2D tab can display a raster image as background (for example orthophoto) behind KOF points and polylines.

Use the top toolbar:

- Open raster...: load raster image (**.jpg**, **.jpeg**, **.png**, **.tif**, **.tiff**, **.bmp**)
- Raster alpha: control transparency
- Clear raster: remove the loaded raster

Raster visibility is controlled in the left Layers panel, where raster appears as its own layer entry.

Georeferencing Requirements

Raster placement uses a world file next to the image. Supported world-file naming:

- **.jgw**, **.jpgw**, **.jpegw** for JPEG
- **.pgw**, **.pngw** for PNG
- **.tfw**, **.tifw**, **.tiffw** for TIFF
- **.bpw**, **.bmpw** for BMP
- **.wld** fallback

Notes:

- Raster and KOF data must be in the same projected CRS for proper alignment.
- Rotated world files (non-zero rotation terms) are currently not supported.
- Very large rasters are automatically loaded as a downsampled preview for faster display; georeferencing still uses full-resolution world-file metadata.

Boundary Tightness

The Boundary tightness control filters long triangulation edges (relative to the median edge length).

- Lower values: tighter concave boundary (drops more outer triangles)
- Higher values: looser boundary (keeps more triangles)

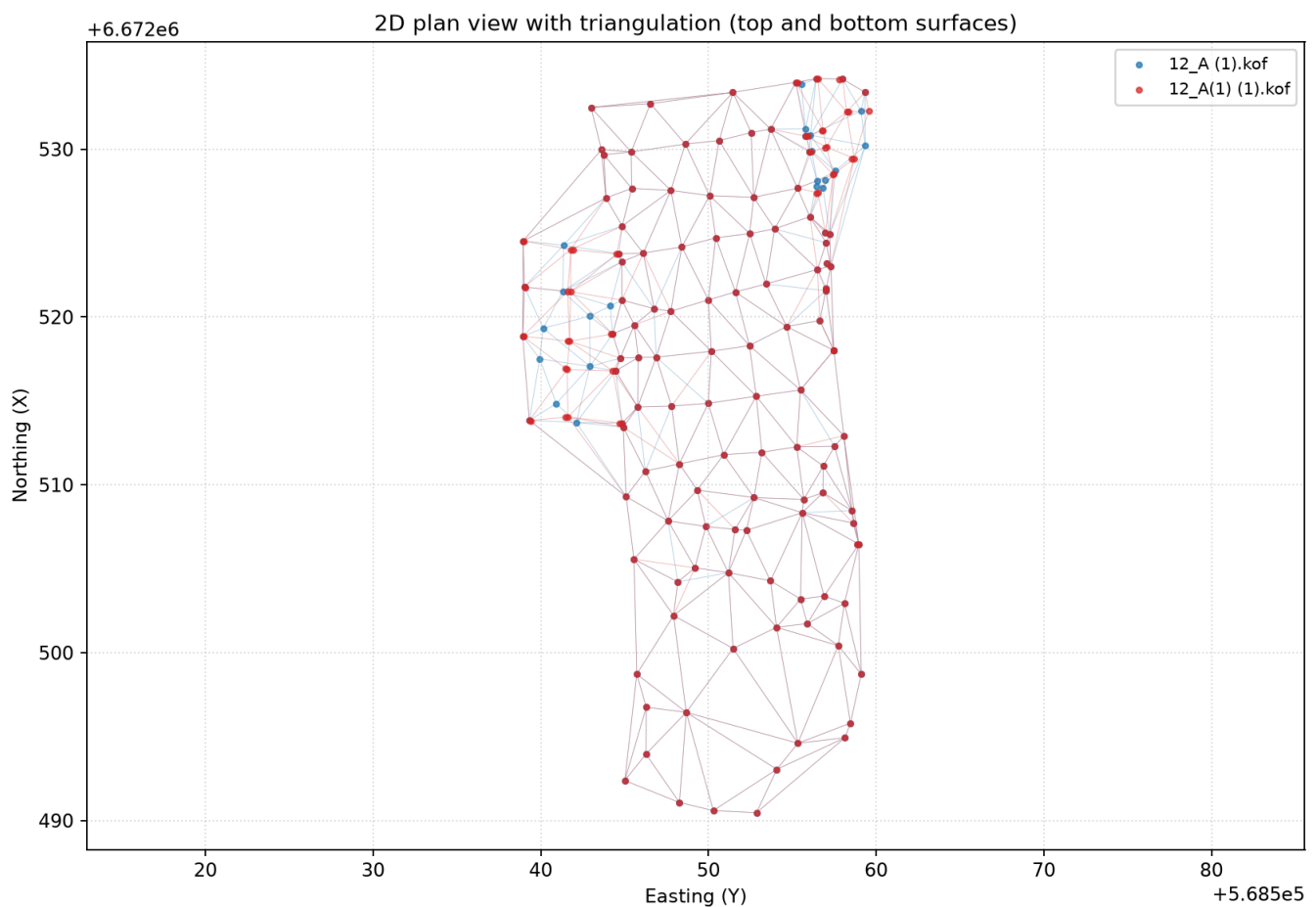
If you change this value after computing volume, the app marks the result as stale and you should recompute.

Example Images

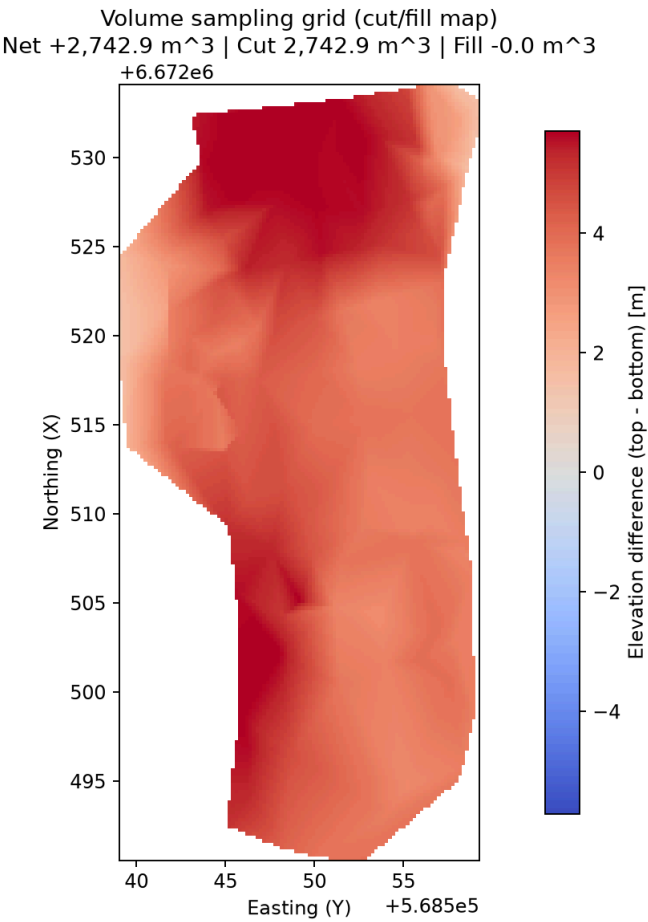
These images are generated from the included sample files:

- 12_A (1).kof
- 12_A(1) (1).kof

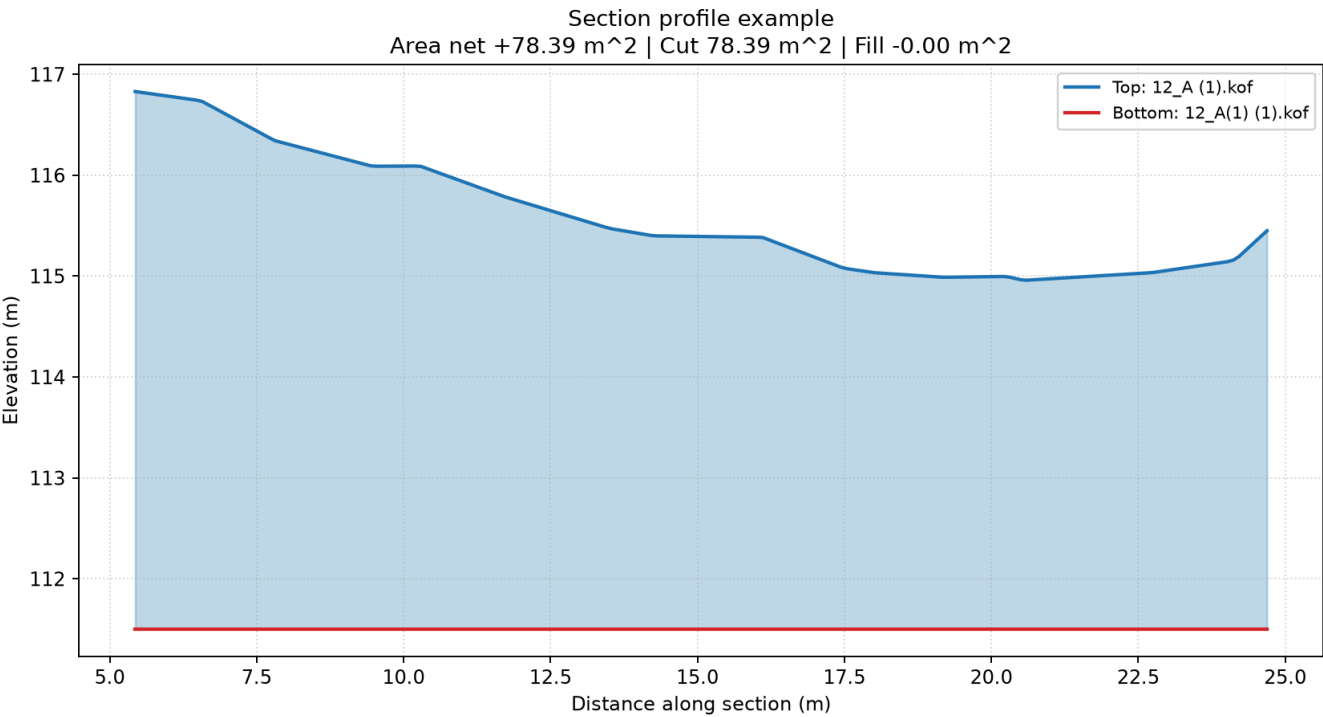
2D Plan + Triangulation



Volume Sampling (Cut/Fill Map)



Section Profile Example



Regenerate README Images

A helper script is included to regenerate the images:

```
python scripts/generate_readme_images.py
```

It writes PNG files to `docs/images/`.

Notes

- Coordinates are interpreted as:
 - `X` = Northing
 - `Y` = Easting
 - `H` = Elevation
- Volume method options:
 - `raster` (default): grid sampling over overlapping extents of the triangulated surfaces.
 - `triangles`: direct integration over overlapping triangle polygons from both surfaces.
- Interpolation outside the filtered triangulation boundary is ignored.